

Theoretical Bounds and Mathematical Mechanisms for Neural Scaling Laws:

A Cross-Disciplinary Mathematical Survey

Theoretical Machine Learning & Information Theory Synthesis

August 15, 2026

Abstract

Empirical scaling laws in deep learning report that held-out cross-entropy loss falls predictably as a power law in model parameters N , training tokens D , and compute $C \approx 6ND$, and are commonly fitted with the additive form $L(N, D) = E + AN^{-\alpha_N} + BD^{-\alpha_D}$. Over the past decade, work in information theory, statistical physics, approximation theory, and high-dimensional statistics has produced a family of candidate mathematical mechanisms for this behaviour. This monograph is a self-contained, rigorous, and deliberately accessible synthesis of that literature.

We review, in increasing order of specificity to deep learning: **(i) rate–distortion theory**, showing why neural scaling is a lossy-compression phenomenon governed by the distortion–rate function of a heavy-tailed source ($\mathcal{D}(R) \asymp R^{-\rho}$) rather than by a channel-coding error exponent; **(ii) minimax nonparametric statistics**, where the Yang–Barron metric-entropy balance fixes the achievable data exponent, and where the Rissanen/Clarke–Barron redundancy bound already contains a N/D trade-off of Chinchilla type; **(iii) language statistics**, including Hilberg’s conjecture and the Cagnetta–Raventós–Ganguli–Wyart horizon-limited prediction $\alpha_D = \gamma/(2\beta)$, the first parameter-free prediction of a scaling exponent from measurable corpus statistics; **(iv) geometry and approximation theory**, covering manifold tiling ($N^{-4/d}$), the Bahri et al. variance/resolution phase diagram, and the impossibility results (DeVore–Howard–Micchelli widths, statistical-query and cryptographic hardness) that bound what any scaling law can achieve; **(v) solvable kernel and linear models**, where exact joint (N, D) laws such as $\Theta(N^{-(b-1)} + D^{-(b-1)/a})$ can be derived and where the algorithm — one-pass SGD versus multi-pass or ridge regression — provably changes the data exponent; and **(vi) discrete mechanisms**: Hutter’s memorizer, skill quantization, associative memories, and superposition.

Three features distinguish this treatment. We supply a cross-disciplinary “Rosetta Stone” and a unified notation, so that the same mechanism can be recognised under five different names. We grade every claim on an explicit four-level evidence scale, separating theorems from solvable models, mechanistic proposals, and curve fits. Finally, we quantify the “honest gap”: no theory yet derives the empirical exponents of natural language from first principles, and we show by explicit numerical substitution which theories are, and are not, mutually consistent with the published Kaplan and Chinchilla fits.

Contents

1	Introduction: How to Read This Monograph	3
1.1	The question	3
1.2	Who this is for, and how it is organised	3
1.3	Notation	3
1.4	A cross-disciplinary Rosetta Stone	5
1.5	Hierarchy of scientific evidence	5
2	What Empirical Scaling Laws Establish in Practice	6
2.1	The measured phenomenon	6
2.2	The compute-optimal allocation rule	6
2.3	Two distinctions that prevent most misreadings	7
3	Information-Theoretic Foundations and Shannon Limits	8
3.1	Cross-entropy decomposition and the irreducible floor	8
3.2	Why scaling is rate-distortion, not a channel reliability exponent	9
3.3	Metric entropy and Yang-Barron minimax rates	10
3.4	Universal coding redundancy: Chinchilla’s ancestor	11
3.5	Information-theoretic model/data allocation: Jeon and Van Roy	12
4	Statistics of Language and Horizon-Limited Scaling	12
4.1	Zipf, Hilberg, and long-range correlations	12
4.2	Deriving the data exponent: Cagnetta, Raventós, Ganguli, and Wyart	13
4.3	A caution from controlled data: random-walk graphs	14
5	Geometry, Approximation, and Fundamental Limits	14
5.1	Manifold tiling: Sharma and Kaplan	14
5.2	Variance-limited versus resolution-limited scaling: Bahri et al.	15
5.3	Impossibility results: what no amount of scaling can buy	16
6	Solvable Models: Kernels, Random Features, and Linear Regression	17
6.1	The cutoff calculus	17
6.2	Exact joint (N, D) laws in linear regression	18
6.3	Dynamics, phases, and finite width	19
7	Discrete Mechanisms: Memorization, Skills, and Superposition	19
7.1	Hutter’s countable-input memorizer	19
7.2	The quantization model of neural scaling	20
7.3	Associative memories with heavy-tailed inputs	20
7.4	Superposition: representing more features than dimensions	20
8	Comparative Synthesis and Consistency Checks	21
8.1	Master taxonomy	21
8.2	Do the theories survive contact with the published numbers?	21
9	The Honest Gap and Open Frontiers	23
9.1	Open frontiers	23
10	Conclusion	24
A	Glossary for Cross-Disciplinary Readers	25
B	Cheat Sheet: Cutoffs and Exponents	26

1 Introduction: How to Read This Monograph

1.1 The question

Train a sequence of transformers of increasing size on increasing amounts of text, plot the held-out cross-entropy loss against model size on log-log axes, and the points fall on a straight line over several orders of magnitude. That is the empirical observation. This monograph is about a single question:

Why a straight line, and what determines its slope?

The question matters practically — the slope decides how to spend a training budget — and theoretically, because a power law is a strong hint about the structure of the underlying data. Exponential decay would tell us the problem is effectively finite-dimensional; a power law tells us it is not.

1.2 Who this is for, and how it is organised

The relevant literature spans at least five research communities that rarely read one another. An information theorist writes $\mathcal{D}(R)$ for the same object a statistician calls a minimax risk, a physicist calls a learning curve, and an approximation theorist calls a width. We therefore assume no single reader is expert in all of them, and each technical section opens with a short primer stating only what is needed.

- **Section 2** states what is actually measured, with numbers, and the practical consequences (compute-optimal allocation) and pitfalls (metric artefacts, broken laws).
- **Section 3** gives the information-theoretic foundations: the irreducible loss floor, rate-distortion scaling, minimax rates, and universal-coding redundancy.
- **Section 4** treats the statistics of language itself, and the one theory that currently predicts a measured exponent with no free parameters.
- **Section 5** covers geometry, approximation theory, and hard impossibility results.
- **Section 6** covers exactly solvable kernel, random-feature, and linear models, where joint (N, D) laws can be proved.
- **Section 7** covers discrete mechanisms: memorization, skills, and superposition.
- **Sections 8–9** synthesise, test the theories numerically against published fits, and state what remains open.
- **Appendix A** is a glossary of terms used across fields, and **Appendix B** is a one-page “resolution rule” cheat sheet.

A reader who wants the shortest path to the main idea can read [Section 2](#), the boxed takeaway in [Section 3.2](#), [Section 4.2](#), and [Section 8](#).

1.3 Notation

Symbols are used consistently throughout; where a source paper uses a conflicting convention we say so explicitly.

Sym.	Meaning	Sym.	Meaning
N	model size (non-embedding parameters)	λ_k	k -th eigenvalue of a covariance or kernel operator
D	dataset size in tokens or samples	a	spectral decay exponent, $\lambda_k \asymp k^{-a}$
C	training compute in FLOPs, $C \approx 6ND$	c_k^2	power of the target in eigenmode k
L	expected cross-entropy loss (nats/token)	b	target decay exponent, $c_k^2 \asymp k^{-b}$
E	irreducible loss floor of a fit	κ	effective ridge / regularization level
h	entropy rate of the source process	k_*	spectral cutoff: modes below it are learned
$\mathcal{E}$	excess (reducible) loss, $L - E$	d	intrinsic dimension of the data manifold
α_N, α_D	parameter and data exponents	s	smoothness (Sobolev order) of the target
α_C	compute exponent of the optimal frontier	ε	resolution / covering radius
R	description rate in bits	n	context length in tokens (Sec. 4)
$\mathcal{D}(R)$	distortion-rate function	γ	conditional-entropy decay exponent
ρ	source spectral tail exponent, $\lambda_k \asymp k^{-(1+\rho)}$	β	token-correlation decay exponent
θ	reverse water-filling threshold	δ	Hilberg exponent, $\gamma = 1 - \delta$
ν	crossover sharpness in joint fits	ζ	Zipf exponent of skills or inputs
M	number of represented features	d_{model}	residual-stream / activation width

Table 1: Notation used throughout. We write $f \asymp g$ when $c_1 g \leq f \leq c_2 g$ for constants $0 < c_1 \leq c_2$ independent of the scaling variable, and $f \sim g$ for informal proportionality.

1.4 A cross-disciplinary Rosetta Stone

Table 2 translates the same four concepts across the four dialects in which the scaling literature is written. Nearly every theory in this monograph is an instance of one pattern: *rank a countable set of primitives by importance, note that the importances have a heavy tail, and determine how many of them a given resource can resolve.*

Concept	Information Theory	Geometry / Approximation	Statistical Physics / ML
Data complexity	Covariance spectrum $\lambda_k \asymp k^{-(1+\rho)}$; metric entropy $H(\varepsilon) \asymp \varepsilon^{-d/s}$	Manifold intrinsic dimension d ; Sobolev smoothness s	Kernel/NTK spectrum $\lambda_k \asymp k^{-a}$; target power $c_k^2 \asymp k^{-b}$
Model capacity	Description rate R bits ($\approx N$ times bits per parameter)	Tiling cells $K \asymp N$; linear regions; covering number	Parameter count N ; width w ; mode cutoff k_*
Data budget	Blocklength; sample count D	Sample covering radius $\varepsilon \asymp D^{-1/d}$	Tokens D ; gradient steps; SGD time
Limiting loss	Distortion-rate $\mathcal{D}(R) \asymp R^{-\rho}$; entropy rate h	Piecewise-Taylor error $N^{-4/d}$; minimax rate $D^{-\frac{2s}{2s+d}}$	Generalization error $\mathcal{E}(D) \asymp D^{-(b-1)}$; Bayes risk E

Table 2: Cross-disciplinary “Rosetta Stone” translating scaling concepts across mathematical fields.

1.5 Hierarchy of scientific evidence

Claims in this literature differ enormously in strength, and conflating them is the main source of confusion. We label every substantive statement with one of four levels, abbreviated [L1]–[L4]:

1. **[L1] Rigorous theorem.** A closed-form result with complete proof for a fully specified source, function class, loss, and algorithm. Examples: Gaussian reverse water-filling; DeVore–Howard–Micchelli n -widths; Yang–Barron minimax rates.
2. **[L2] Controlled solvable model.** An exact or asymptotic result in a stylized but well-defined model: random features, high-dimensional ridge regression, infinite-width NTK, one-pass SGD on linear regression, associative memories.
3. **[L3] Mechanistic proposal.** A geometric or physical mechanism transferred from a solvable model to real trained networks, with testable assumptions: manifold tiling, Zipfian skill quantization, horizon-limited context truncation.
4. **[L4] Empirical observation.** A parametric curve fitted to training runs over a finite resource window.

The central honest fact of the field, developed in Section 9: every quantitative prediction of a real language-model exponent currently requires at least one [L3] step, and every [L1] statement about real data requires empirically measured inputs.

2 What Empirical Scaling Laws Establish in Practice

2.1 The measured phenomenon

Let N denote model capacity (conventionally the non-embedding parameter count), D the number of training tokens processed, and C the training compute in FLOPs. A ubiquitous empirical finding [L4] is that validation loss decays smoothly over several orders of magnitude and is well fitted by

$$L(N, D) = E + \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{D_c}{D}\right)^{\alpha_D} = E + AN^{-\alpha_N} + BD^{-\alpha_D}, \quad (1)$$

with $A = N_c^{\alpha_N}$, $B = D_c^{\alpha_D}$. Here E is the fitted irreducible floor, interpreted as the Bayes risk of the data distribution under log loss, and $\alpha_N, \alpha_D > 0$ are the scaling exponents.

The compute proxy is

$$C \approx 6ND, \quad (2)$$

which counts roughly $2ND$ FLOPs for the forward pass and $4ND$ for the backward pass over non-embedding parameters [45]. It ignores the $O(n_{\text{ctx}})$ attention term (negligible while $d_{\text{model}} \gg n_{\text{ctx}}/12$), and it must be modified for mixture-of-experts models, where active and total parameters differ [22, 47].

Equation (1) was documented across modalities by Hestness et al. [36] and Rosenfeld et al. [64], extended to many generative modalities by Henighan et al. [35], and made canonical for language models by Kaplan et al. [45] and Hoffmann et al. (Chinchilla) [38]. Table 3 collects the principal published fits.

Study	Setting	α_N	α_D	Notes
Kaplan et al. [45]	WebText2, GPT-2 class	0.076	0.095	Joint fit uses a non-additive form and sets $E = 0$; $L(C_{\min}) \propto C^{-0.050}$.
Hoffmann et al. [38]	MassiveText, Chinchilla	0.34	0.28	Approach 3 fit: $E = 1.69$, $A = 406.4$, $B = 410.7$ nats/token.
Cagnetta et al. [18]	TinyStories, GPT-2/LLaMA	–	≈ 0.19	Data-limited regime; predicted from corpus statistics.
Cagnetta et al. [18]	WikiText	–	≈ 0.14	As above.

Table 3: Representative published exponents. The Kaplan and Hoffmann exponents differ by a factor of 3–4; Section 2.2 explains why, and Section 8.2 shows how much theoretical inference depends on which one is used.

Caution: the additive fit is an ansatz, not a theorem

Equation (1) asserts asserts that the parameter bottleneck and the data bottleneck do not interact: the excess loss is a sum, not a joint function. No theory derives this in general, and the solvable models of Section 6 reproduce additivity only in particular regimes. Kaplan et al. themselves used a non-additive joint form, and smoothed variants (Section 5.2) fit better near the crossover. Treat α_N and α_D as parameters of a specific fitted functional form over a specific measurement window, not as universal constants of nature.

2.2 The compute-optimal allocation rule

Given a compute budget $C \approx 6ND$, how should it be split between N and D ? Minimize (1) subject to $6ND = C$. Writing the Lagrangian $\mathcal{L} = E + AN^{-\alpha_N} + BD^{-\alpha_D} + \mu(6ND - C)$, setting

$\partial_N \mathcal{L} = \partial_D \mathcal{L} = 0$ and multiplying the first condition by N and the second by D eliminates μ and gives the stationarity condition

$$\alpha_N A N^{-\alpha_N} = \alpha_D B D^{-\alpha_D}, \quad (3)$$

i.e. *at the optimum the two loss terms are in fixed ratio*. Substituting $D = C/(6N)$ into (3) and solving:

$$\boxed{N_{\text{opt}}(C) \propto C^{\frac{\alpha_D}{\alpha_N + \alpha_D}}, \quad D_{\text{opt}}(C) \propto C^{\frac{\alpha_N}{\alpha_N + \alpha_D}}.} \quad (4)$$

Substituting back gives the loss along the optimal frontier, a result worth isolating because it is the exponent that actually governs progress per dollar:

$$\boxed{L(C) - E \propto C^{-\alpha_C}, \quad \alpha_C = \frac{\alpha_N \alpha_D}{\alpha_N + \alpha_D}.} \quad (5)$$

Thus α_C is half the harmonic mean of α_N and α_D : it is dominated by whichever exponent is *smaller*. Improving the larger exponent buys almost nothing.

Numbers. For the Chinchilla fit ($\alpha_N = 0.34$, $\alpha_D = 0.28$), (4) gives $N_{\text{opt}} \propto C^{0.45}$ and $D_{\text{opt}} \propto C^{0.55}$ (Hoffmann et al. report 0.46 and 0.54 from their direct fit), so parameters and tokens should scale in near-equal proportion, $D/N \approx 20$ tokens per parameter at the scales they studied. The compute exponent is $\alpha_C = (0.34)(0.28)/0.62 = 0.154$: a $10\times$ increase in compute reduces the reducible loss by a factor $10^{-0.154} \approx 0.70$, i.e. by 30%. For the Kaplan fit, $\alpha_C = (0.076)(0.095)/0.171 = 0.042$, close to their directly reported $L(C_{\text{min}}) \propto C^{-0.050}$.

Why Kaplan and Chinchilla disagreed. Kaplan et al. reported $N_{\text{opt}} \propto C^{0.73}$, $D_{\text{opt}} \propto C^{0.27}$ — parameters growing far faster than data. Note that this does *not* follow from their own α_N, α_D via (4), which would give $C^{0.56}$: the discrepancy arises because they optimized a different object (loss at fixed step count with early stopping, rather than the converged additive fit). Besiroglu et al. [10] re-examined the Chinchilla regression and found the reported confidence intervals implausibly tight and the third-approach fit sensitive to estimation details. Porian et al. [61] then isolated three concrete causes of the Kaplan–Chinchilla gap: counting embedding parameters in N , tuning the learning-rate schedule and warmup independently of scale, and the choice of optimizer hyperparameters. Correcting these reconciles the two datasets, and the modern consensus favours the near-linear Chinchilla relation.

The optimum is not the deployment optimum. Equation (4) minimizes loss per unit *training* compute. If a model will serve many inference requests, the total-cost optimum shifts toward smaller models trained on more tokens than Chinchilla prescribes [66]. Most deployed “overtrained” models are rational under that objective, not violations of the scaling law.

2.3 Two distinctions that prevent most misreadings

- **Smooth loss versus discontinuous capability metrics.** Pretraining cross-entropy improves smoothly, but downstream accuracy on discrete tasks often appears to jump at a threshold — “emergent abilities” [80]. Schaeffer, Miranda, and Koyejo [67] showed [**L4, with a clean analytical model**] that such jumps can be produced by the *metric*: an exact-match or 0/1 accuracy applies a sharp nonlinearity to a smoothly improving per-token log-probability, and the same runs scored with a continuous metric (edit distance, Brier score, average log-probability) improve smoothly. This is a demonstration that thresholded metrics are *sufficient* to create apparent emergence, not a proof that no genuine phase transitions occur; solvable models do exhibit real emergence in the underlying computation [54, 58], and higher-resolution evaluations can make some capability curves predictable

in advance [4, 39]. The practical rule is to fit scaling laws to smooth quantities and to treat any claimed discontinuity as a claim about a metric until shown otherwise.

- **Single versus broken power laws.** Fitted laws are valid over the measured window. Caballero et al. [15] introduced broken power laws with smooth transitions between regimes, which capture plateaus, the effects of repeated data [56], and saturation toward E . Alabdulmohsin et al. [2] make a similar point with a functional form that interpolates between a random-guess plateau and the asymptote, and emphasize how badly exponents extrapolate when estimated from too narrow a window. Observational scaling laws fitted across many existing public models [65] provide a cheaper and broader alternative to training one’s own ladder.

3 Information-Theoretic Foundations and Shannon Limits

Primer: the two Shannon theories, and which one applies

Shannon’s theory has two halves. *Channel coding* asks how reliably one can transmit through noise, and its answers are exponential in blocklength. *Rate–distortion* (lossy source coding) asks how well one can describe a source using R bits, and its answers can be polynomial in R when the source has infinitely many, slowly decaying degrees of freedom. Neural scaling is the second kind of problem: the parameters are a lossy description of the data distribution. That single observation explains why we see power laws rather than exponentials.

3.1 Cross-entropy decomposition and the irreducible floor

Let P^* be a stationary ergodic ground-truth process on token sequences, $Z_t = (X_1, \dots, X_t)$ the context, and $Q(\cdot | Z_t)$ the model’s predictive distribution. Then [L1]

$$\mathbb{E}_{P^*}[-\log Q(X_{t+1} | Z_t)] = H_{P^*}(X_{t+1} | Z_t) + \mathbb{E}_{Z_t}[D_{\text{KL}}(P^*(\cdot | Z_t) \| Q(\cdot | Z_t))]. \quad (6)$$

Since the Kullback–Leibler divergence is non-negative and vanishes only when $Q \equiv P^*$, the conditional entropy is an unbeatable floor for log loss, and *all* of the reducible loss is a KL divergence. This is the reason log loss is the natural object of a scaling theory: the quantity being scaled away is exactly an information divergence.

For a stationary process on a finite vocabulary, $H(X_{t+1} | X_1^t)$ is non-increasing in t and converges to the entropy rate [24, 70]

$$h = \lim_{t \rightarrow \infty} H(X_{t+1} | X_1^t) = \lim_{t \rightarrow \infty} \frac{1}{t} H(X_1, \dots, X_t), \quad (7)$$

so the fitted floor obeys $E \geq h$, with equality only for a model with unbounded context. Any finite-context model has a strictly larger floor — a fact that Section 4 turns into a scaling law.

Worked conversion: is a frontier model near the Shannon limit?

Comparing model losses to Shannon’s estimate of the entropy of English requires two conversions, both of which are frequently botched.

1. **Nats to bits:** $1 \text{ nat} = 1/\ln 2 = 1.443 \text{ bits}$.
2. **Tokens to characters:** modern byte-pair encoders average roughly 4 characters per token on English prose; the exact figure is tokenizer- and corpus-dependent.

The Chinchilla irreducible floor is $E = 1.69 \text{ nats/token}$ [38]. That is $1.69 \times 1.443 = 2.44 \text{ bits/token}$, or $\approx 0.61 \text{ bits/character}$ at 4 characters per token. Shannon’s 1951 human letter-guessing experiment [71] bracketed printed English at 0.6–1.3 bits/character; compression-based extrapolations of large corpora put the entropy rate near 1 bit/character [77]. So the fitted floor of a frontier-scale language model sits at or just below the bottom of Shannon’s band.

Three caveats keep this from being a clean “we have reached the Shannon limit” claim. (i) E is a fitted extrapolation parameter of (1), not a measured entropy, and it is strongly correlated with A, B, α_N, α_D in the regression [10]. (ii) Web-scale training corpora are not the literary English of Shannon’s experiment; they contain boilerplate, code, and near-duplicates, all of which lower measured entropy per character. (iii) Shannon’s estimate is itself an upper bound derived from human predictors, and there is no reason to think human next-letter prediction is optimal. The honest statement is that model and human-derived estimates of the entropy rate of English now agree to within a factor of about two.

3.2 Why scaling is rate–distortion, not a channel reliability exponent

A recurring intuition tries to link neural scaling to Shannon’s channel-coding error exponent. The analogy fails, and seeing why is instructive.

Error exponents versus distortion–rate

Channel coding: transmitting at rate $R < C_{\text{chan}}$ over a memoryless channel with block-length n , the decoding error probability decays *exponentially*, $P_e(n) \leq e^{-nE_r(R)}$.

Neural scaling: the excess loss decays *polynomially*, $L(N) - E \asymp N^{-\alpha_N}$.

The structural difference: in channel coding, n independent uses of the same channel produce exponential concentration of measure. In neural scaling, N parameters constitute a *single* description of a fixed, infinite-dimensional object. Nothing is repeated, so nothing concentrates exponentially. What N buys is resolution, and the return on resolution is set by how fast the source’s importance spectrum decays.

Theorem 3.1 (Gaussian reverse water-filling on a power-law spectrum; **L1**). Let X be a zero-mean Gaussian random element of a separable Hilbert space with covariance eigenvalues

$$\lambda_k = c k^{-(1+\rho)}, \quad c > 0, \rho > 0, k \in \mathbb{N}^+, \quad (8)$$

so that $\sum_k \lambda_k < \infty$. Under squared-error distortion $\mathcal{D} = \mathbb{E}\|X - \hat{X}\|^2$, the distortion–rate function satisfies, as $R \rightarrow \infty$,

$$\boxed{\mathcal{D}(R) \asymp R^{-\rho}}. \quad (9)$$

Proof. By reverse water-filling for parallel Gaussian sources [9, 24], the rate–distortion curve is parameterized by a threshold $\theta > 0$:

$$\mathcal{D}(\theta) = \sum_{k \geq 1} \min(\theta, \lambda_k), \quad R(\theta) = \frac{1}{2} \sum_{k \geq 1} \log_2 \max\left(1, \frac{\lambda_k}{\theta}\right). \quad (10)$$

Only modes with $\lambda_k > \theta$ receive rate. Their number is

$$K(\theta) = \max\{k : ck^{-(1+\rho)} \geq \theta\} = \left\lfloor (c/\theta)^{\frac{1}{1+\rho}} \right\rfloor \asymp \theta^{-\frac{1}{1+\rho}}. \quad (11)$$

For $k \leq K$ we have $\lambda_k/\theta \approx (K/k)^{1+\rho}$, so

$$\begin{aligned} R(\theta) &= \frac{1+\rho}{2} \sum_{k=1}^K \log_2 \frac{K}{k} = \frac{1+\rho}{2} \log_2 \frac{K^K}{K!} \\ &= \frac{1+\rho}{2 \ln 2} \left[K \ln K - (K \ln K - K + \mathcal{O}(\ln K)) \right] = \frac{(1+\rho) \log_2 e}{2} K(\theta) + \mathcal{O}(\log K), \end{aligned} \quad (12)$$

using Stirling's approximation. Hence the rate is *proportional to the number of active modes*: $R \asymp K \asymp \theta^{-1/(1+\rho)}$, equivalently $\theta(R) \asymp R^{-(1+\rho)}$.

The distortion is the noise injected into active modes plus the discarded tail:

$$\mathcal{D}(\theta) = K(\theta)\theta + \sum_{k>K(\theta)} ck^{-(1+\rho)} \approx K\theta + \int_K^\infty cx^{-(1+\rho)} dx = K\theta + \frac{c}{\rho} K^{-\rho}. \quad (13)$$

Substituting $K = (c/\theta)^{1/(1+\rho)}$, both terms have the same order:

$$\mathcal{D}(\theta) \asymp \left(1 + \frac{1}{\rho}\right) c^{\frac{1}{1+\rho}} \theta^{\frac{\rho}{1+\rho}}. \quad (14)$$

Finally $\theta(R) \asymp R^{-(1+\rho)}$ gives $\mathcal{D}(R) \asymp (R^{-(1+\rho)})^{\rho/(1+\rho)} = R^{-\rho}$. $\square$

Corollary 3.2 (Finite-dimensional sources decay exponentially; **L1**). If $\lambda_k = \lambda$ for $k \leq d$ and $\lambda_k = 0$ for $k > d$, then at high rate all d modes are active, $R = \frac{d}{2} \log_2(\lambda/\theta)$ and $\mathcal{D} = d\theta$, giving

$$\mathcal{D}(R) = d\lambda 2^{-2R/d} = d\lambda \exp\left(-\frac{2 \ln 2}{d} R\right). \quad (15)$$

Remark 3.1 (Closing the loop to α_N). Theorem 3.1 concerns rate R in bits, not parameters N . Two further steps [**L3**] are needed to reach $L(N)$. First, one must posit $R \asymp N$ (each parameter carries $\Theta(1)$ useful bits, up to the precision floor discussed in [48]). Second, one must identify excess log loss with squared distortion; this holds to leading order because KL divergence is locally quadratic in the parameter perturbation, $D_{\text{KL}}(P\|P+\Delta) = \frac{1}{2} \Delta^\top I \Delta + \mathcal{O}(\|\Delta\|^3)$. Granting both, $\alpha_N = \rho$: *the parameter exponent equals the tail exponent of the source spectrum*. Neither step is innocuous, and both are places where a factor of 2 can enter.

Foundational takeaway: the spectral fingerprint

Power-law scaling $L(N) - E \asymp N^{-\alpha}$ is the signature of an effectively infinite-dimensional, heavy-tailed source. Had natural data been genuinely finite-dimensional, optimal compression would give exponential improvement and the entire scaling-law industry would not exist. Conversely, any intervention that truncates the tail — deduplication, pruning, curriculum — can in principle change the functional form, not just the constants (Section 9).

3.3 Metric entropy and Yang–Barron minimax rates

Rate–distortion bounds what a description can achieve; minimax theory bounds what an *estimator* can achieve from D samples. The two are linked by the same object: the metric entropy of the model class.

Definition 3.1 (Metric entropy). For a class $\mathcal{P}$ with metric $\mathbf{d}$, the covering number $\mathcal{N}(\varepsilon, \mathcal{P}, \mathbf{d})$ is the smallest number of ε -balls covering $\mathcal{P}$, and the metric entropy is $H(\varepsilon; \mathcal{P}) = \log \mathcal{N}(\varepsilon, \mathcal{P}, \mathbf{d})$. It is the number of bits needed to specify an element of $\mathcal{P}$ to accuracy ε .

Theorem 3.3 (Yang–Barron balance; **L1** [78,81]). Let $\mathcal{P}$ be a class of densities whose Hellinger metric entropy satisfies $H(\varepsilon; \mathcal{P}) \asymp \varepsilon^{-r}$, $r > 0$. Then the minimax risk over D i.i.d. samples satisfies

$$\inf_{\hat{p}} \sup_{p^* \in \mathcal{P}} \mathbb{E}_{p^*} [\mathbf{d}^2(p^*, \hat{p})] \asymp \varepsilon_D^2, \quad (16)$$

where ε_D solves the balance equation

$$\boxed{H(\varepsilon_D; \mathcal{P}) \asymp D \varepsilon_D^2}. \quad (17)$$

Reading the balance equation

In (17), the left side is the number of nats needed to name the truth to accuracy ε ; the right side is the number of nats of information D samples supply about a target at that accuracy. The optimal accuracy is where supply meets demand. Every classical nonparametric rate is this one equation with a different entropy.

Remark 3.2 (Technical caveat). The upper bound in (17) uses the *global* metric entropy (via a Bayes/mixture argument with a covering prior); the matching lower bound is proved by Fano’s inequality applied to a *local* packing, and equality of the two entropies is a regularity property of the class. For the standard smoothness classes below they coincide; for irregular classes they need not, and the balance equation can be off. The result also bounds a minimax risk over a class — it says nothing about the rate for one particular truth, which is where the extra structure of real data enters.

Corollary 3.4 (Sobolev rates; **L1**). For densities in a Sobolev ball $W^s([0, 1]^d)$, classical entropy estimates give $H(\varepsilon) \asymp \varepsilon^{-d/s}$. Solving $\varepsilon_D^{-d/s} \asymp D \varepsilon_D^2$ yields $\varepsilon_D^{2+d/s} \asymp D^{-1}$, hence

$$\mathcal{R}_D \asymp D^{-\frac{2s}{2s+d}} \implies \alpha_D = \frac{2s}{2s+d}. \quad (18)$$

This is the curse of dimensionality in its statistical form: for fixed smoothness s , the exponent collapses as $1/d$ for large d . It is also the first quantitative bridge to the geometric picture: if the data lie on a d -dimensional manifold rather than filling the ambient space, d in this formula is the *intrinsic* dimension, which is how deep networks evade the ambient curse [57,68].

3.4 Universal coding redundancy: Chinchilla’s ancestor

An older literature already contains an N -versus- D trade-off, in exactly the form scaling laws rediscovered.

Theorem 3.5 (Parametric redundancy; Rissanen, Clarke–Barron; **L1** [23,62,63]). For a regular, identifiable k -parameter family $\{p_\vartheta : \vartheta \in \Theta \subset \mathbb{R}^k\}$ with Fisher information $I(\vartheta)$, the asymptotic minimax redundancy of universal coding over D observations is

$$\boxed{\inf_q \sup_{\vartheta \in \Theta} \mathbb{E}_{p_\vartheta} \left[\log \frac{p_\vartheta(X^D)}{q(X^D)} \right] = \frac{k}{2} \log \frac{D}{2\pi e} + \log \int_\Theta \sqrt{\det I(\vartheta)} d\vartheta + o(1)}. \quad (19)$$

Remark 3.3 (Why this looks like Chinchilla). Divide by D to obtain excess loss *per token*: a k -parameter model costs $\frac{k}{2D} \log D + O(1/D)$ nats/token in estimation error, on top of whatever approximation error remains. Writing $k = N$ and adding an approximation term $AN^{-\alpha_N}$ gives

$$L(N, D) - E \approx \underbrace{AN^{-\alpha_N}}_{\text{approximation}} + \underbrace{\frac{N \log D}{2D}}_{\text{estimation}}, \quad (20)$$

which, minimized under $ND \propto C$, again produces a near-linear optimal relation between N and D up to logarithmic factors. The Chinchilla prescription is thus visible in a 1986 theorem, provided one is willing to treat a transformer as a regular parametric family — which it emphatically is not (it is unidentifiable, over-parameterized, and its Fisher information is singular). The singular-learning-theory correction replaces $k/2$ by the real log-canonical threshold, which is generally smaller.

3.5 Information-theoretic model/data allocation: Jeon and Van Roy

Jeon and Van Roy [43] make the preceding heuristic rigorous without assuming regularity, using a Bayesian rate-distortion argument [L1/L2]. For a data-generating process F and a compressed surrogate $\tilde{F}$, the *cumulative* excess log loss over the first T observations obeys

$$\frac{1}{T} \sum_{t < T} \mathcal{E}_t \leq \underbrace{\frac{\mathbb{I}(F; \tilde{F})}{T}}_{\text{estimation / information cost}} + \underbrace{\mathbb{E}[D_{\text{KL}}(P_F(\cdot | X) \| P_{\tilde{F}}(\cdot | X))]}_{\text{misspecification cost}}. \quad (21)$$

The first term is the price of identifying $\tilde{F}$ from data and falls as $1/T$; the second is the structural error of the surrogate and falls as capacity grows. The bound is on the *average* excess loss over the first T observations (an online-to-batch statement), not on the loss of the final model — a distinction worth keeping, since the two coincide only when the learning curve is regular.

Instantiating this for data generated by an infinite-width two-layer neural network and optimizing (21) under a compute constraint $C \propto NT$, they conclude that *the optimal relation between model size and data size is linear up to logarithmic factors*, an information-theoretic derivation of the Chinchilla rule that does not depend on fitting a curve. A companion analysis treats compute-optimal scaling directly [42].

4 Statistics of Language and Horizon-Limited Scaling

The theories of Section 3 say that exponents are determined by properties of the source. This section is about the one domain where those properties have been *measured* and turned into a prediction.

4.1 Zipf, Hilberg, and long-range correlations

Three empirical regularities of natural language matter here.

Zipf’s law. Word frequencies decay as a power law in rank, $p_k \propto k^{-\zeta}$ with $\zeta \approx 1$ [83]. This alone guarantees a heavy-tailed set of primitives to be learned, and drives the discrete models of Section 7.

Long-range correlations. Correlations between tokens decay as a power law in separation rather than exponentially [33, 50]; an exponentially mixing source (a finite-order Markov chain) would give exponentially small returns to context length, contradicting observation.

Hilberg’s conjecture. In 1990 Hilberg [37], reinterpreting Shannon’s data, conjectured that the block entropy of natural language grows sub-linearly, $H(X_1^n) \propto n^\delta$ with $\delta \approx 0.5$ — which as stated implies a *zero* entropy rate. The form used today is Dębowski’s *relaxed* Hilberg conjecture [27, 28], which permits a positive rate:

$$H(X_1^n) = h n + c n^\delta + o(n^\delta), \quad 0 < \delta < 1. \quad (22)$$

Since $H(X_{n+1} | X_1^n) = H(X_1^{n+1}) - H(X_1^n)$ exactly, differencing (22) gives (assuming the remainder differences behave, which requires a regularity assumption not implied by (22) alone)

$$H(X_{n+1} | X_1^n) - h \asymp c \delta n^{-\gamma}, \quad \gamma = 1 - \delta. \quad (23)$$

This is the key structural fact. A model restricted to n tokens of context cannot beat $H(X_{n+1} | X_1^n)$, so its floor exceeds the true entropy rate by $\asymp n^{-\gamma}$. Any mechanism that limits usable context therefore generates a power law. Hilberg’s $\delta \approx 0.5$ corresponds to $\gamma \approx 0.5$; the directly measured values below are $\gamma \approx 0.27\text{--}0.34$, i.e. $\delta \approx 0.66\text{--}0.73$, so the modern measurements are qualitatively but not quantitatively consistent with Hilberg’s original figure.

4.2 Deriving the data exponent: Cagnetta, Raventós, Ganguli, and Wyart

Cagnetta, Raventós, Ganguli, and Wyart [18] connect two measurable statistics of a corpus to the data-limited scaling exponent, with no fitted parameters and no synthetic data model. This is currently the strongest quantitative link between corpus statistics and a neural scaling exponent.

Conceptual intuition: reading through sampling noise

1. Correlations between tokens separated by n positions decay as $\|\mathbf{C}(n)\| \asymp n^{-\beta}$.
2. Estimating such a correlation from D tokens carries central-limit sampling noise of order $D^{-1/2}$.
3. A correlation is only *learnable* if it exceeds the noise floor: $n^{-\beta} \gtrsim D^{-1/2}$. This defines a statistically resolvable context horizon

$$n^*(D) \asymp D^{\frac{1}{2\beta}}.$$

4. If the model masters everything inside that horizon, its remaining excess loss is the conditional-entropy gap at context length n^* , which by (23) is

$$\mathcal{L}(D) - H_\infty \asymp (n^*(D))^{-\gamma} = D^{-\frac{\gamma}{2\beta}}.$$

In words: *more data does not mainly make the model better at short-range prediction; it makes the model able to see further.*

Theorem 4.1 (Horizon-limited data scaling; **L3** [18]). Suppose a corpus satisfies:

1. **Conditional entropy decay:** $H_n - H_\infty \asymp n^{-\gamma}$, with $H_n = H(X_{t+1} | X_{t-n+1}^t)$;
2. **Token correlation decay:** $\|\mathbf{C}(n)\|_{\text{op}} \asymp n^{-\beta}$ for the token–token correlation matrix at lag n ;
3. **Fast within-horizon learning:** structure at scales $n \ll n^*(D)$ is essentially acquired, so the binding constraint is the horizon itself, not within-horizon approximation.

Then in the data-limited regime,

$$\boxed{\mathcal{L}(D) - H_\infty \asymp D^{-\alpha_D}, \quad \alpha_D = \frac{\gamma}{2\beta}}. \quad (24)$$

Parameter-free empirical test. The authors measure γ and β directly on corpora and compare $\gamma/(2\beta)$ with the exponents of training curves for GPT-2- and LLaMA-style transformers trained from scratch:

$$\textbf{TinyStories: } \gamma = 0.34, \beta = 0.88 \implies \alpha_D^{\text{pred}} = 0.34/(2 \times 0.88) \approx 0.19, \quad (25)$$

$$\textbf{WikiText: } \gamma = 0.27, \beta = 0.94 \implies \alpha_D^{\text{pred}} = 0.27/(2 \times 0.94) \approx 0.14, \quad (26)$$

matching the measured data-limited exponents on the two benchmarks. The theory also predicts a data collapse: loss curves measured at different context lengths n superimpose under the rescaling $D \mapsto D/n^{2\beta}$, $\mathcal{L}_n \mapsto n^\gamma \mathcal{L}_n$, which is a sharper test than matching a single slope.

What this does and does not settle

It is a genuine first: a scaling exponent predicted from measurable corpus statistics with no free parameters. But (i) it predicts α_D only, in the *data-limited* regime, and says nothing about α_N ; (ii) assumption 3 is a strong dynamical claim about optimization, not a property of the corpus, and is the step that keeps this at evidence level **[L3]**; (iii) it explains α_D in terms of γ and β , which are themselves unexplained — the “why” has been moved one level down, not eliminated.

4.3 A caution from controlled data: random-walk graphs

Barkeshli, Alfarano, and Gromov [7] trained transformers on sequences generated by random walks (bigram processes) on graphs of tunable complexity, and by progressively simplified generative language models. They find neural scaling laws *even when the data have no power-law correlation structure at all*, and track how exponents evolve as the generative process is simplified from a 4-layer transformer down to a bigram model.

The implication is important and cuts against a naive reading of the whole preceding section: power-law structure in the data is *sufficient* but apparently not *necessary* for power-law learning curves over a finite range. Optimization dynamics can manufacture effective power laws on their own. Any claim of the form “the exponent is X because the data statistic is Y ” therefore needs a control showing that the relationship fails when Y is changed.

5 Geometry, Approximation, and Fundamental Limits

Primer: why dimension is the villain

To approximate a function to accuracy ε on a d -dimensional domain, one generally needs about ε^{-d} pieces of information: the number of ε -balls needed to cover the domain grows exponentially in d . Invert this and every rate in this section appears: with N pieces one gets resolution $\varepsilon \asymp N^{-1/d}$, and the error is whatever the local approximation leaves over at that resolution. The whole game is (i) what d really is for natural data, and (ii) what power of ε the local error carries.

5.1 Manifold tiling: Sharma and Kaplan

High-dimensional data typically concentrate near a low-dimensional submanifold $\mathcal{M}$ of intrinsic dimension d far below the ambient dimension [3, 49, 60]. Sharma and Kaplan [72] model a network as a piecewise-affine approximator on $\mathcal{M}$ — reasonable for ReLU networks, whose number of linear regions grows with depth and width [55] — and deduce the parameter exponent.

Geometric intuition: flat tiles on a curved surface

1. N parameters buy $K \asymp N$ tiles, each of linear size $\varepsilon \asymp N^{-1/d}$.
 2. A $\mathcal{C}^2$ target curves away from a flat tile by its second-order Taylor remainder: $|f^*(x) - \hat{f}(x)| \leq \frac{1}{2} \|x - x_0\|^2 \|\nabla^2 f^*\| \asymp \varepsilon^2$.
 3. Squared error integrates the square of that: $L(N) \asymp \varepsilon^4 = N^{-4/d}$.
- The “4” is two factors of 2: one from second-order Taylor, one from squaring.

Theorem 5.1 (Manifold tiling exponent; **L3** [72]). Let $\mathcal{M}$ be a compact d -dimensional Riemannian manifold and $f^* \in \mathcal{C}^2(\mathcal{M})$. If a network with N parameters induces $K \asymp N$ local linear regions of diameter $\varepsilon \asymp N^{-1/d}$, and the regions are placed so as to resolve $\mathcal{M}$ uniformly, then under squared error

$$L(N) - E \asymp N^{-4/d} \implies \alpha_N = \frac{4}{d}. \quad (27)$$

If the target is merely Lipschitz, the local error is $\mathcal{O}(\varepsilon)$ rather than $\mathcal{O}(\varepsilon^2)$ and $\alpha_N = 2/d$. The same exponents apply to cross-entropy loss because KL divergence is locally quadratic in the predictive perturbation.

Measured intrinsic dimensions, and an honest mismatch. Sharma and Kaplan tested $\alpha \approx 4/d$ on image classifiers, where it fares reasonably, and on GPT-type language models, where it does not. For GPT-2 small the empirical $\alpha_N \approx 0.076$ implies $4/\alpha_N \approx 53$, whereas their nearest-neighbour estimate of the intrinsic dimension of the token activations gives $d \geq 90$ across heterogeneous documents — but only $d \approx 7$ within a single contiguous 1024-token passage. Their own conclusion is that “the data manifold has a scale-dependent structure, and may not be well-characterized by a single intrinsic dimension.” We stress this because the mismatch is often omitted when the $4/d$ law is quoted. Note also that the inferred d inherits all of the fragility of α_N : using the Chinchilla exponent $\alpha_N = 0.34$ instead of Kaplan’s 0.076 changes the implied dimension from 53 to 12 (Section 8.2).

5.2 Variance-limited versus resolution-limited scaling: Bahri et al.

Bahri, Dyer, Kaplan, Lee, and Sharma [6] organize the mechanisms into a phase diagram with four regimes [**L2/L3**], according to whether the bottleneck is statistical fluctuation (variance-limited) or insufficient resolution of the data manifold (resolution-limited), along the data axis or the model axis.

Two conclusions of that paper deserve emphasis. First, the variance-limited exponents are *universal* — $1/D$ and $1/w$, independent of the data — whereas the resolution-limited exponents are data-dependent; observing a non-trivial exponent is thus already evidence that one is in a resolution-limited regime. Second, in the random-feature/kernel setting with a power-law spectrum they derive a *duality*: the model-resolution and data-resolution exponents coincide and equal the spectral decay exponent, $\alpha_N = \alpha_D = \alpha_K$. Empirically $\alpha_N \neq \alpha_D$ for language models, which is itself informative: it says the two axes are not being limited by the same mechanism (Section 6.2 gives a mechanism where they provably differ).

Across the crossover between regimes, the additive fit (1) is usually replaced by a smoothed form

$$L(N, D) = E + \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\nu}} + \left(\frac{D_c}{D} \right)^{\frac{\alpha_D}{\nu}} \right]^\nu, \quad \nu > 0, \quad (28)$$

which recovers (1) at $\nu = 1$ and sharpens the transition as $\nu \rightarrow 0$.

Regime	Limiting mechanism	Asymptotic excess loss
Data, variance-limited	Sample fluctuations about the infinite-data limit	$\Delta L \propto D^{-1}$ (central limit theorem)
Width, variance-limited	Fluctuations about the infinite-width (GP/NTK) limit	$\Delta L \propto w^{-1}$
Data, resolution-limited	Finite sample coverage of a d -manifold	$\Delta L \propto D^{-n/d}$, with $n \geq 4$ for an accurate piecewise-linear fit and a smooth loss
Model, resolution-limited	Finite parameter coverage of a d -manifold	$\Delta L \propto N^{-n/d}$

Table 4: The four scaling regimes of Bahri et al. [6]. The exponent n in the resolution-limited rows is the order of the leading non-vanishing term in the local error expansion: $n = 1$ for a Lipschitz bound on the loss, $n = 2$ for a smooth loss with a first-order fit, and $n \geq 4$ when the model is an accurate piecewise-linear approximant and the loss has a quadratic minimum.

5.3 Impossibility results: what no amount of scaling can buy

Scaling laws describe what happens on natural data. It is equally important to know what is impossible in general, because those bounds tell us how much of the observed success is a property of the data rather than of the architecture.

Theorem 5.2 (Continuous nonlinear n -width; DeVore–Howard–Micchelli 1989; **L1** [31]). Let $W_p^s([0, 1]^d)$ be a Sobolev ball and let $\mathcal{M}_N = \{g(\vartheta, \cdot) : \vartheta \in \mathbb{R}^N\}$ be any N -parameter family in which the parameter selection $f \mapsto \vartheta(f)$ is *continuous*. Then, under standard conditions on p, q , the approximation error in L_q obeys

$$\boxed{d_N(W_p^s([0, 1]^d))_q \geq c N^{-s/d}}. \quad (29)$$

Remark 5.1 (Read the hypothesis carefully). The continuity of $f \mapsto \vartheta(f)$ is essential, not decorative. Yarotsky [82] showed that very deep ReLU networks attain the strictly faster rate $N^{-2s/d}$ — twice the exponent — using a weight assignment that is necessarily *discontinuous* in f (bits of the target are encoded in the low-order bits of weights). So the correct statement is not “gradient descent cannot beat $N^{-s/d}$ ”, but “any method whose parameters depend continuously on the target cannot”. Whether trained networks behave like continuous or discontinuous selectors is an open empirical question. See DeVore, Hanin, and Petrova [30] for a modern survey of the whole landscape.

Remark 5.2 (Escaping the curse requires structure, and there are several exits). The $N^{-s/d}$ barrier applies to worst-case Sobolev functions. Real targets are not worst case, and at least three exits are known: (i) *Barron classes*, functions with integrable Fourier-weighted spectra, are approximated at the dimension-free rate $\mathcal{O}(N^{-1/2})$ [5, 8]; (ii) *low intrinsic dimension*, where d is replaced by the manifold dimension [57]; (iii) *compositional structure*, where a hierarchical target is learned hierarchically and the effective dimension is that of each local factor [17, 68]. All three are claims about data, and all three are consistent with the empirical fact that scaling works far better than worst-case theory allows.

Theorem 5.3 (Statistical-query hardness of parities; **L1** [11, 46]). The statistical-query dimension of the class of parity functions on d bits is 2^d . Consequently any SQ algorithm using queries of tolerance τ requires at least $\tau^2 2^d - 1$ queries; taking $\tau = 2^{-d/3}$ gives $2^{\Omega(d)}$ queries. Since minibatch gradient descent with bounded precision is (approximately) an SQ algorithm, this bounds gradient-based learning of such targets.

Remark 5.3 (Hardness of *learning*, not of representation). A two-layer network represents a parity easily; what is hard is *finding* it from data. Under standard cryptographic assumptions there exist function classes that no polynomial-time algorithm can learn to non-trivial advantage even though small networks represent them [26, 69]. This is the sharpest sense in which scaling laws are contingent: on adversarially structured data, no exponent exists at all. That we observe clean power laws on natural language is a statement about natural language.

Takeaway: scaling laws are statements about data

$L \asymp N^{-\alpha}$ is not a law of neural networks. It is a joint property of (i) the low intrinsic dimension, heavy-tailed spectrum, and compositional hierarchy of natural data, and (ii) the resolution rule of the training algorithm. Change either — adversarial data, or a different optimizer — and the exponent changes or disappears.

6 Solvable Models: Kernels, Random Features, and Linear Regression

This section contains the strongest results in the survey: models simple enough that joint (N, D) scaling laws can be *proved*, yet rich enough to display the phenomenology.

Primer: spectral bias in one paragraph

Diagonalize the kernel: any target decomposes into eigenmodes ordered by eigenvalue $\lambda_1 \geq \lambda_2 \geq \dots$, like a Fourier series ordered from low to high frequency. Gradient-based learning fits high-eigenvalue modes first — this is *spectral bias*. Every learning curve in this section then has the same two-line derivation: (1) determine the cutoff k_* , the number of modes the resource resolves; (2) the error is the target power left unlearned, $\sum_{k > k_*} c_k^2 \asymp k_*^{-(b-1)}$. Different resources and algorithms differ *only* in step (1).

6.1 The cutoff calculus

In the infinite-width Neural Tangent Kernel regime [21, 41], training is kernel regression with a fixed kernel. Let the kernel integral operator have eigenvalues $\lambda_k \asymp k^{-a}$ ($a > 1$) and let the target’s power in eigenmode k be $c_k^2 \asymp k^{-b}$ ($b > 1$), so $\sum_k c_k^2 < \infty$. Bordelon, Canatar, and Pehlevan [13, 19] and Spigler, Geiger, and Wyart [76] established the spectral-bias picture; the residual error above a cutoff k_* is

$$\mathcal{E} \approx \sum_{k > k_*} c_k^2 \asymp \int_{k_*}^{\infty} x^{-b} dx \asymp k_*^{-(b-1)}. \quad (30)$$

Cutoff 1: ridgeless kernel regression on D samples. The effective ridge κ solves the self-consistent equation $D \asymp \sum_k \frac{\lambda_k}{\lambda_k + \kappa}$. With $\lambda_k \asymp k^{-a}$, $\sum_k (1 + \kappa k^a)^{-1} \asymp \kappa^{-1/a}$, so $\kappa \asymp D^{-a}$; modes are resolved while $\lambda_k \gtrsim \kappa$, giving $k_* \asymp \kappa^{-1/a} \asymp D$. Hence, in the noiseless case,

$$\mathcal{E}(D) \asymp D^{-(b-1)}. \quad (31)$$

Intuitively, D samples resolve D modes — the most favourable possible rule.

Cutoff 2: label noise with optimal ridge. With label noise of variance σ^2 , keeping k_* modes contributes variance $\asymp \sigma^2 k_*/D$; balancing this against the bias $k_*^{-(b-1)}$ gives $k_* \asymp D^{1/b}$ and

$$\mathcal{E}(D) \asymp D^{-\frac{b-1}{b}}, \quad (32)$$

the classical optimal rate, established in this power-law parameterization by Cui et al. [25] and, in the classical capacity/source notation, by Caponnetto and De Vito [20]. Note $\frac{b-1}{b} < 1$ always: noise caps the achievable data exponent below 1 regardless of how fast the target decays.

Cutoff 3: one-pass SGD. One pass of SGD does not invert the covariance. Mode k accumulates gradient signal at rate λ_k per step, so after D steps it is learned only if $D\lambda_k \gtrsim 1$, giving the much stingier rule $k_* \asymp D^{1/a}$ and

$$\mathcal{E}(D) \asymp D^{-\frac{b-1}{a}}. \quad (33)$$

Caution: saturation, and a notational trap that has caused real confusion

Saturation. Equation (31) holds while the target is rough enough to be the binding constraint; for very smooth targets the rate saturates and stops improving with b (the classical KRR saturation effect [25]). Rates quoted without a validity window should be distrusted.

Convention. Two incompatible conventions for “target decay” coexist in this literature. We use $c_k^2 = \text{power of the target in the } L^2\text{-orthonormal eigenmode } k$, so $\|f\|^2 = \sum_k c_k^2$. Many physics-flavoured papers instead write $f = \sum_k \vartheta_k \sqrt{\lambda_k} \phi_k$ with $\vartheta_k^2 \asymp k^{-b'}$, i.e. they normalize inside the RKHS. The two are related by $b = a + b'$. This is exactly why the same one-pass-SGD result appears in the literature both as $D^{-(b-1)/a}$ and as $D^{-\frac{a+b'-1}{a}}$: they are the same formula in different clothes. Always check which normalization a quoted exponent assumes before comparing it to another paper’s.

6.2 Exact joint (N, D) laws in linear regression

Lin, Wu, Kakade, Bartlett, and Lee [52] prove a genuine two-variable scaling law [L2].

Theorem 6.1 (Lin et al., NeurIPS 2024 [52]). Consider infinite-dimensional linear regression in which the model is a linear function of N sketched covariates, trained by one-pass SGD on D samples. Assume the covariate covariance has a power-law spectrum $\lambda_k \asymp k^{-a}$ with $a > 1$, and that the optimal parameter follows a Gaussian prior (which makes the target power $c_k^2 \asymp \lambda_k$, i.e. $b = a$). Then the reducible part of the test error is

$$\mathcal{E}(N, D) = \Theta\left(N^{-(a-1)} + D^{-\frac{a-1}{a}}\right). \quad (34)$$

Moreover the variance term, which grows with N , is dominated by the other terms because of the implicit regularization of SGD — so test error is monotonically improving in model size, as scaling laws require and as classical bias–variance reasoning would not predict.

Writing this in the general- b notation of Section 6.1, the two mechanisms are the two cutoff rules:

$$\mathcal{E}(N, D) \asymp \underbrace{N^{-(b-1)}}_{\text{sketch truncates at } k_*=N} + \underbrace{D^{-\frac{b-1}{a}}}_{\text{one-pass SGD reaches } k_*=D^{1/a}}, \quad (35)$$

and the general- b statement (together with the multi-pass extension) is established by Lin, Wu, and Bartlett [51].

Why the model and data exponents differ — and a testable prediction

Both terms come from the same formula $\mathcal{E} \asymp k_*^{-(b-1)}$; only the cutoff differs.

- **Parameters:** an N -dimensional sketch simply discards modes beyond $k_* = N$. Exponent $b - 1$.
- **Data:** one pass of SGD reaches only $k_* = D^{1/a}$. Exponent $(b - 1)/a$.

Hence the model states a relation between the two empirical exponents:

$$\alpha_N = b - 1, \quad \alpha_D = \frac{b - 1}{a} \quad \implies \quad \boxed{\alpha_N = a \alpha_D}, \quad a > 1.$$

Because $a > 1$ is required for a summable spectrum, the model *predicts* $\alpha_N > \alpha_D$. This is a falsifiable statement, and Section 8.2 checks it against both published fits — where it passes for Chinchilla and fails for Kaplan.

Remark 6.1 (Data reuse interpolates between the two extremes). Multi-pass SGD with L iterations over D samples achieves $\Theta(N^{1-b} + L^{-(b-1)/a})$ with $L \lesssim D^{a/b}$ [51]. At the maximum useful number of passes the data exponent becomes $D^{-\frac{b-1}{b}}$ — exactly the optimally-regularized kernel-regression rate (32). So the algorithm interpolates continuously between the one-pass exponent $(b - 1)/a$ and the statistically optimal $(b - 1)/b$. *The data exponent is not a property of the data alone; it is a property of the data and the optimizer.* This is one of the cleanest theoretical results in the field and it directly explains why epoching, batch size, and learning-rate schedules move measured exponents.

6.3 Dynamics, phases, and finite width

Maloney, Roberts, and Sully [53] give a solvable random-feature model reproducing joint (N, D) laws and their compute-optimal frontier, with the spectrum of the feature covariance controlling everything. Bordelon, Atanasov, and Pehlevan [12] extend the analysis to gradient-descent *dynamics*, deriving loss-versus-time curves and explaining why early-time and late-time exponents differ and how finite width enters. Paquette, Paquette, Xiao, and Pennington [59] map the full compute-optimal phase diagram, proving the existence of 4+3 distinct phases depending on the relative values of the data spectrum and target decay exponents — with different phases obeying different compute-optimal allocation rules, so that even (4) is regime-dependent. Defilippis, Loureiro, and Misiakiewicz [29] give dimension-free deterministic equivalents for random-feature ridge regression, sharpening these rates and identifying the minimum number of features needed to reach the minimax rate. Earlier high-dimensional analyses of generalization dynamics by Advani, Saxe, and Sompolinsky [1] contain much of the same mathematics in the guise of double descent.

7 Discrete Mechanisms: Memorization, Skills, and Superposition

The models so far are continuous. But much of what a language model knows is discrete: facts, idioms, rare words, individual skills. Remarkably, discrete models produce the same power laws, from the same underlying cause — a heavy-tailed ranking.

7.1 Hutter’s countable-input memorizer

The simplest model in the entire field [L1]. Hutter [40] considers an optimal memorizer over countably many discrete inputs with probabilities $\theta_i \propto i^{-(1+\eta)}$, $\eta > 0$. After D samples, an

input has likely been seen if $\theta_i \gtrsim 1/D$, giving $i_* \asymp D^{1/(1+\eta)}$; everything else is unseen and contributes its probability mass:

$$\mathbb{E}[\mathcal{E}_D] = \sum_{i>i_*} \theta_i \asymp i_*^{-\eta} = \boxed{D^{-\frac{\eta}{1+\eta}}}. \quad (36)$$

No optimization, no geometry, no network: a power-law input distribution plus a coverage argument suffices. The lesson is that observing a power law is very weak evidence for any particular mechanism.

7.2 The quantization model of neural scaling

Michaud, Liu, Girit, and Tegmark [54] propose [L3] that knowledge decomposes into a discrete set of independent “quanta” — sub-problems, facts, or skills — learned in order of decreasing use frequency.

Theorem 7.1 (Quantization scaling; L3 [54]). Let quantum i have use frequency $U_i \asymp i^{-\zeta}$ with $\zeta > 1$ (Zipf’s law for skills), let each quantum cost $\Theta(1)$ parameters, and let quanta be acquired greedily in frequency order. A network with N parameters masters the top $K \asymp N$ quanta, leaving

$$L(N) - E = \sum_{i>N} U_i \approx \int_N^\infty x^{-\zeta} dx = \frac{N^{-(\zeta-1)}}{\zeta-1} \implies \boxed{\alpha_N = \zeta - 1}. \quad (37)$$

If instead a quantum requires τ examples to be acquired, it is learned once $D U_i \gtrsim \tau$, so $i_* \asymp D^{1/\zeta}$ and

$$\boxed{\alpha_D = \frac{\zeta - 1}{\zeta} = \frac{\alpha_N}{\alpha_N + 1}}. \quad (38)$$

Two observations. First, the data exponent coincides exactly with Hutter’s — the quantization model is Hutter’s memorizer with skills in place of inputs. Second, the relation $\alpha_D = \alpha_N/(\alpha_N + 1)$ is a sharp, parameter-free prediction linking the two measured exponents, and it is testable: Chinchilla’s $\alpha_N = 0.34$ predicts $\alpha_D = 0.25$ against a measured 0.28 — close; Kaplan’s $\alpha_N = 0.076$ predicts $\alpha_D = 0.071$ against a measured 0.095. The model also offers an account of emergence: individual quanta appear abruptly even though their frequency-weighted sum is smooth.

Nam et al. [58] make this exactly solvable in the multitask sparse-parity setting, obtaining closed forms for emergence times and for scaling in time, data, model size, and compute simultaneously.

7.3 Associative memories with heavy-tailed inputs

Cabannes, Dohmatob, and Bietti [16] analyze outer-product associative memories storing discrete key–value associations drawn from Zipf-distributed inputs, and derive exact sample- and parameter-scaling curves [L2], including how the exponents depend on the memory scheme and the optimization algorithm. This closes part of the gap between the purely combinatorial models above and gradient-trained networks: the storage capacity of a d -dimensional memory is finite, and the interaction between capacity and a Zipfian demand distribution reproduces power-law loss curves from first principles.

7.4 Superposition: representing more features than dimensions

How can a residual stream of width d_{model} represent $M \gg d_{\text{model}}$ semantic features? Elhage et al. [34] showed empirically and analytically that networks use *superposition*: features are assigned nearly-orthogonal directions and tolerated interference, which is viable precisely because the features are sparse.

Proposition 7.2 (Interference noise under superposition; **L2**). Let $\mathbf{W} \in \mathbb{R}^{d_{\text{model}} \times M}$ have unit-norm columns. By the Johnson–Lindenstrauss lemma [44], there exist such frames with pairwise coherence $|\langle w_i, w_j \rangle| \leq \varepsilon$ whenever $d_{\text{model}} = \mathcal{O}(\varepsilon^{-2} \log M)$; a random frame achieves $\mathbb{E} \langle w_i, w_j \rangle^2 = 1/d_{\text{model}}$. Let features x_j be independent, each zero-mean with variance s^2 when active and active with probability p . Linear read-out $\hat{x}_i = w_i^\top (\mathbf{W}x) = x_i + \sum_{j \neq i} \langle w_i, w_j \rangle x_j$ then incurs cross-talk of variance

$$\mathbb{E} \left[\left(\sum_{j \neq i} \langle w_i, w_j \rangle x_j \right)^2 \right] = p s^2 \sum_{j \neq i} \langle w_i, w_j \rangle^2 \approx \boxed{p s^2 \frac{M}{d_{\text{model}}}}. \quad (39)$$

So interference falls as $1/d_{\text{model}}$ at fixed M : a network can afford to represent $M \asymp d_{\text{model}}/p$ features before cross-talk overwhelms signal. On its own this yields a $1/d_{\text{model}}$ law, not a non-trivial exponent; a genuine power law requires combining it with a heavy-tailed distribution of feature *importances*, at which point the mechanism becomes a continuous analogue of the quantization model: widening the network admits more of the ranked feature list, and the loss tracks the unrepresented tail. Superposition also supplies a mechanistic reason why capability curves can look sharp: a feature is either allocated a direction or not.

8 Comparative Synthesis and Consistency Checks

8.1 Master taxonomy

Table 5 summarises the frameworks. Read down the “derived exponent” column and one pattern is unmistakable.

The unifying pattern: rank, then resolve

Every entry has the same two-factor structure:

$$\text{scaling exponent} = (\text{tail decay of ranked primitives}) \times (\text{algorithmic resolution rule}).$$

The primitives may be covariance eigenmodes, kernel modes, manifold cells, Zipfian skills, discrete inputs, or context scales; the resolution rule may be sketch truncation ($k_* = N$), sample coverage ($k_* = D$), gradient accumulation ($k_* = D^{1/a}$), noise-limited horizon ($n^* = D^{1/2\beta}$), or frequency coverage ($i_* = D^{1/\zeta}$). A power law appears whenever a heavy-tailed hierarchy is resolved progressively. This is also why the observation of a power law identifies neither the primitives nor the rule: *many mechanisms produce the same phenomenology, and distinguishing them requires measuring the inputs independently* — which is precisely what makes [18] and [7] methodologically important.

8.2 Do the theories survive contact with the published numbers?

A survey that only catalogues theories is not doing its job. Each framework maps measured exponents onto data statistics, so each can be checked for internal consistency against the two canonical fits. Table 6 does this.

Three lessons. (i) *Theories that predict a relation between α_N and α_D are worth more than theories that predict either alone*, because they can fail. (ii) The Kaplan exponent pair is inconsistent with the one-pass-SGD model ($a < 1$ would make the covariance spectrum non-summable), independent evidence that the Kaplan fit reflected under-tuned training rather than data structure — exactly what Porian et al. [61] concluded from a different direction. (iii) Inferences *about data* drawn from fitted exponents (“the manifold dimension of language is 53”) are only as stable as the fit, and the fits have moved by factors of four.

Framework	Core assumptions	Machinery	Derived exponent	References	Lvl
Rate-distortion	Gaussian source, spectrum $\lambda_k \asymp k^{-(1+\rho)}$	Reverse water-filling, high-rate asymptotics	$\mathcal{D}(R) \asymp R^{-\rho}$; $\alpha_N = \rho$ if $R \asymp N$	[9, 70]	L1
Metric entropy / minimax	Sobolev class $W^s([0, 1]^d)$, $H(\varepsilon) \asymp \varepsilon^{-d/s}$	Yang–Barron balance $H(\varepsilon_D) \asymp D\varepsilon_D^2$; Fano	$\alpha_D = \frac{2s}{2s+d}$	[78, 81]	L1
Universal coding / MDL	Regular k -parameter family	Rissanen–Clarke–Barron redundancy	$\mathcal{E} \asymp \frac{N \log D}{2D}$; $N \propto D$	[23, 43, 63]	L1
Language horizon	$\ C(n)\ \asymp n^{-\beta}$, $H_n - h \asymp n^{-\gamma}$	Sampling-noise horizon $n^* \asymp D^{1/(2\beta)}$	$\alpha_D = \frac{\gamma}{2\beta}$	[18, 27, 37]	L3
Manifold tiling	Data on d -manifold, $\mathcal{C}^2$ target	Voronoi tiling, 2nd-order Taylor remainder	$\alpha_N = \frac{4}{d}$ ($\frac{2}{d}$ if Lipschitz)	[72]	L3
Variance and resolution phases	Finite N , D , width w	Manifold covering + CLT fluctuations	D^{-1} , w^{-1} , $D^{-n/d}$, $N^{-n/d}$	[6]	L2/L3
Kernel spectra	$\lambda_k \asymp k^{-a}$, $c_k^2 \asymp k^{-b}$	Spectral bias; effective ridge $\kappa \asymp D^{-a}$	$D^{-(b-1)}$ noiseless; $D^{-\frac{b-1}{b}}$ noisy	[13, 19, 20, 25]	L1/L2
Joint (N, D) linear	$\lambda_k \asymp k^{-a}$, Gaussian prior, one-pass SGD	Sketch truncation + accumulated gradient SNR	$\Theta(N^{-(b-1)} + D^{-\frac{b-1}{a}})$	[51–53, 59]	L2
Memorization	Inputs Zipf, $\theta_i \propto i^{-(1+\eta)}$	Coverage of unseen mass	$\alpha_D = \frac{\eta}{1+\eta}$	[40]	L1
Skill quantization	Skills with Zipf utilities $U_i \asymp i^{-\zeta}$	Greedy allocation, tail integration	$\alpha_N = \zeta - 1$, $\alpha_D = \frac{\zeta-1}{\zeta}$	[16, 54, 58]	L3
Superposition	$M \gg d_{\text{model}}$ sparse features	Johnson–Lindenstrauss, cross-talk variance	Noise $\propto \frac{pM}{d_{\text{model}}}$	[34, 44]	L2

Table 5: Master taxonomy of theoretical frameworks for neural scaling laws, with evidence level from Section 1.5.

Theory and its map	Kaplan fit (0.076, 0.095)	Chinchilla fit (0.34, 0.28)	Verdict
Manifold tiling: $d = 4/\alpha_N$	$d \approx 53$	$d \approx 12$	Inference about data geometry swings by 4× with the fit.
One-pass SGD: $\alpha_N = a \alpha_D$, needs $a > 1$	$a = 0.80$ — inadmissible	$a = 1.21$, $b = 1.34$; admissible ($1 < b < a + 1$)	The model <i>falsifies</i> the Kaplan pair; consistent with Chinchilla.
Quantization: $\alpha_D = \alpha_N/(\alpha_N + 1)$	predicts 0.071 vs. 0.095 measured	predicts 0.254 vs. 0.280 measured	Right order; both under-predict by ~ 10 –25%.
Kernel (noisy, optimal ridge): $\alpha_D = (b - 1)/b < 1$	$b = 1.105$	$b = 1.389$	Always satisfiable; weak test.
Compute frontier: $\alpha_C = \frac{\alpha_N \alpha_D}{\alpha_N + \alpha_D}$	0.042 vs. reported 0.050	0.154	Internally consistent in both.

Table 6: Numerical consistency of theories with the two canonical published fits. The exercise shows that some frameworks (one-pass SGD, quantization) make genuinely falsifiable predictions relating α_N to α_D , while others merely absorb any measurement into a free parameter.

9 The Honest Gap and Open Frontiers

The fundamental honest gap

No existing theory derives the empirical scaling exponents of a real language model from first principles. What the theories establish is a *reduction*: the exponent is a function of measurable properties of the data and the algorithm,

$$\alpha = \Phi(a_{\text{spectral}}, d_{\text{manifold}}, \zeta_{\text{Zipf}}, \gamma_{\text{entropy}}, \beta_{\text{correlation}}; \text{optimizer}). \quad (40)$$

Those inputs are measured, not derived. The best current result [18] closes the loop for α_D in the data-limited regime by measuring γ and β on the actual corpus — a real advance, and still a reduction rather than an explanation. Explaining *why* human language has $\gamma \approx 0.3$ and $\beta \approx 0.9$ would require a theory of language itself, and is the field’s deepest open problem.

9.1 Open frontiers

1. **Changing the exponent by changing the data.** Sorscher et al. [75] showed within an exactly solvable student–teacher perceptron model that, *given an oracle pruning metric*, error can in principle decay exponentially rather than polynomially in dataset size; empirically, on CIFAR-10, SVHN, and ImageNet, careful pruning gives better-than-power-law scaling, though not the theoretical exponential. This is the most striking demonstration that the exponent is not fixed. A predictive theory of “effective data quality” — covering deduplication, difficulty curricula, and mixture weights — remains open.
2. **Finite data and synthetic data.** Repeating data yields diminishing returns with a quantifiable decay [56], and the stock of public human text is projected to bind in the late

2020s [79]. Training on model-generated data changes the scaling law itself: Dohmatob et al. [32] show that synthetic data truncates the distribution’s tail — precisely the object that produces the power law — yielding loss of scaling and finite-error plateaus, and Shumailov et al. [73] document the same collapse phenomenon empirically. Since every theory in this monograph attributes the power law to a heavy tail, tail truncation is the theoretically predicted failure mode, and it is now observed.

3. **Test-time compute.** Spending compute at inference — search, sampling, verification, chain-of-thought — trades against pretraining scale [14, 74], and success rates under repeated sampling follow their own regular curves. A unified theory relating the rate–distortion cost of a fixed representation to the cost of search at inference does not yet exist, and would be the natural extension of Section 3.
4. **Beyond fixed kernels.** All the exactly solvable models freeze the representation (lazy training [21]). Real networks learn features, which changes both the spectrum and the cutoff rule; the dynamical model of [12] and the compositional analyses of [17] are the current frontier. A feature-learning theory would be the first to explain why α_N and α_D take the values they do rather than inheriting them from an assumed spectrum.
5. **Other resource axes.** Numerical precision [48], sparsity and routing [22, 47], and inference-aware allocation [66] each introduce their own scaling axis interacting with N and D . The rate–distortion picture is the natural language for precision in particular: it fixes the bits-per-parameter constant that Section 3 had to assume.
6. **From loss to capability.** Cross-entropy is predictable; what practitioners want is downstream capability. Progress requires either metrics with enough resolution to be predictable [39], theories linking skill composition to loss [4, 54], or observational laws fitted across model families [65].

10 Conclusion

Neural scaling laws are not accidents of engineering, but neither are they laws of nature. They are the macroscopic signature of a specific mathematical situation: a learner with finite resolution progressively resolving a heavy-tailed hierarchy of structures in data.

Four things can be said with confidence. *First*, the correct Shannon-theoretic frame is rate–distortion, not channel coding: power-law rather than exponential improvement is the fingerprint of an effectively infinite-dimensional source (Theorem 3.1). *Second*, fundamental limits are real and quantitative — Yang–Barron metric entropy, DeVore–Howard–Micchelli widths, Rissanen redundancy, and statistical-query hardness jointly imply that the observed exponents are contingent on natural data being low-dimensional, heavy-tailed, and compositional. *Third*, in solvable models the exponent is provably a joint property of data *and algorithm*: one-pass SGD, multi-pass SGD, and ridge regression give three different data exponents on identical data. *Fourth*, the mechanisms are not mutually exclusive; rate–distortion, manifold tiling, kernel spectra, and skill quantization are largely the same calculation in different vocabularies, which is why they so often agree.

What remains open is the part that matters most. No theory yet explains why natural language has the particular statistics it has, and every prediction of a real exponent still requires those statistics as input. The most promising recent development — predicting α_D from measured correlation and entropy decay with no free parameters — makes the dependence explicit rather than removing it. Meanwhile the empirical ground is shifting under the theory: data is becoming the binding constraint, synthetic data measurably degrades the very tail the theory relies on, and compute is migrating from training to inference. A theory able to predict exponents *and* to say how they change under these interventions is the natural next goal.

A Glossary for Cross-Disciplinary Readers

Bayes risk / irreducible loss

The lowest achievable expected loss, attained by the true conditional distribution. For log loss it equals the conditional entropy of the source; the fitted constant E in (1) is an estimate of it.

Distortion–rate function $\mathcal{D}(R)$

The least achievable expected distortion when a source is described using R bits. The inverse function is the rate–distortion function $R(\mathcal{D})$.

Reverse water-filling

The optimal allocation of bits across independent Gaussian components: components with variance below a threshold θ receive no bits and are simply discarded; the rest are quantized to distortion θ .

Metric entropy / covering number

log of the number of ε -balls needed to cover a function class: the bits required to specify a member to accuracy ε .

Minimax rate

The best worst-case error achievable by any estimator on a class, as a function of sample size. Upper bounds come from constructing an estimator; lower bounds from information-theoretic testing arguments (Fano, Le Cam, Assouad).

n -width

The best achievable worst-case approximation error over a class using an n -parameter family; the “continuous nonlinear” variant additionally requires the parameters to depend continuously on the target.

Neural Tangent Kernel (NTK)

The kernel describing the linearization of a network about its initialization; in the infinite-width limit with standard scaling, training follows kernel regression with this fixed kernel (“lazy training”).

Spectral bias

The empirical and theoretical tendency of gradient-based learning to fit high-eigenvalue (typically smooth, low-frequency) kernel modes before low-eigenvalue ones.

Effective ridge κ

The self-consistent regularization level induced by having only D samples; it sets which modes are resolved, via $\lambda_{k_*} \asymp \kappa$.

Source and capacity exponents

Classical kernel-regression parameters: capacity is the eigenvalue decay rate a ; source is the smoothness of the target relative to the RKHS. Both are encoded here by the pair (a, b) .

Superposition

Representing more features than dimensions by assigning nearly-orthogonal directions and tolerating interference; viable when features are sparse.

Quanta

Hypothesized discrete, independent units of model knowledge or skill, learned in order of use frequency.

Statistical query (SQ) model

A learning model in which the algorithm receives noisy estimates of expectations rather than individual samples; a good abstraction of minibatch gradient descent, and the setting for strong lower bounds.

B Cheat Sheet: Cutoffs and Exponents

Every learning curve in Sections 6–7 follows from two ingredients: a ranked tail and a resolution rule. With target power $c_k^2 \asymp k^{-b}$ and residual $\mathcal{E} \asymp k_*^{-(b-1)}$:

Resource and algorithm	cutoff k_*	excess loss	Source
N -dimensional sketch or truncation	N	$N^{-(b-1)}$	§6.2
Ridgeless kernel regression, D samples, no noise	D	$D^{-(b-1)}$	§6.1
Kernel ridge regression, noisy, optimal ridge	$D^{1/b}$	$D^{-\frac{b-1}{b}}$	§6.1
One-pass SGD, D steps, spectrum k^{-a}	$D^{1/a}$	$D^{-\frac{b-1}{a}}$	§6.2
Multi-pass SGD, $L \lesssim D^{a/b}$ iterations	$L^{1/a}$	$L^{-\frac{b-1}{a}} \rightarrow D^{-\frac{b-1}{b}}$	§6.2
Manifold tiling with N cells, $\mathcal{C}^2$ target	$\varepsilon = N^{-1/d}$	$N^{-4/d}$	§5.1
Zipf inputs/skills $\propto i^{-\zeta}$, D samples	$D^{1/\zeta}$	$D^{-\frac{\zeta-1}{\zeta}}$	§7.1, 7.2
Correlation horizon, noise floor $D^{-1/2}$	$n^* = D^{\frac{1}{2\beta}}$	$D^{-\frac{\gamma}{2\beta}}$	§4.2
Rate-distortion, R bits, spectrum $k^{-(1+\rho)}$	$K \asymp R$	$R^{-\rho}$	§3.2

Table 7: The resolution rules. Note that rows 2–5 share identical data and identical targets and differ only in the algorithm, yet give four different data exponents.

References

- [1] Madhu S. Advani, Andrew M. Saxe, and Haim Sompolinsky. High-Dimensional Dynamics of Generalization Error in Neural Networks. *Neural Networks*, 132:428–446, 2020.
- [2] Ibrahim Alabdulmohsin, Behnam Neyshabur, and Xiaohua Zhai. Revisiting Neural Scaling Laws in Language and Vision. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.
- [3] Alessio Ansuini, Alessandro Laio, Jakob H. Macke, and Davide Zoccolan. Intrinsic Dimension of Data Representations in Deep Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.
- [4] Sanjeev Arora and Anirudh Goyal. A Theory for Emergence of Complex Skills in Language Models. *arXiv preprint arXiv:2307.15936*, 2023.
- [5] Francis Bach. Breaking the Curse of Dimensionality with Convex Neural Networks. *Journal of Machine Learning Research*, 18(19):1–53, 2017.

- [6] Yasaman Bahri, Ethan Dyer, Jared Kaplan, Jaehoon Lee, and Utkarsh Sharma. Explaining Neural Scaling Laws. *Proceedings of the National Academy of Sciences (PNAS)*, 121(27):e2311878121, 2024.
- [7] Maissam Barkeshli, Alberto Alfarano, and Andrey Gromov. On the Origin of Neural Scaling Laws: From Random Graphs to Natural Language. *arXiv preprint arXiv:2601.10684*, 2026.
- [8] Andrew R. Barron. Universal Approximation Bounds for Superpositions of a Sigmoidal Function. *IEEE Transactions on Information Theory*, 39(3):930–945, 1993.
- [9] Toby Berger. *Rate Distortion Theory: A Mathematical Basis for Data Compression*. Prentice-Hall, 1971.
- [10] Tamay Besiroglu, Ege Erdil, Matthew Barnett, and Josh You. Chinchilla Scaling: A Replication Attempt. *arXiv preprint arXiv:2404.10102*, 2024.
- [11] Avrim Blum, Merrick Furst, Jeffrey Jackson, Michael Kearns, Yishay Mansour, and Steven Rudich. Weakly Learning DNF and Characterizing Statistical Query Learning Using Fourier Analysis. In *Proceedings of the 26th Annual ACM Symposium on Theory of Computing (STOC)*, pages 253–262, 1994.
- [12] Blake Bordelon, Alexander Atanasov, and Cengiz Pehlevan. A Dynamical Model of Neural Scaling Laws. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 4345–4382, 2024.
- [13] Blake Bordelon, Abdulkadir Canatar, and Cengiz Pehlevan. Spectrum Dependent Learning Curves in Kernel Regression and Wide Neural Networks. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *Proceedings of Machine Learning Research*, pages 1024–1034, 2020.
- [14] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large Language Monkeys: Scaling Inference Compute with Repeated Sampling. *arXiv preprint arXiv:2407.21787*, 2024.
- [15] Ethan Caballero, Kshitij Gupta, Irina Rish, and David Krueger. Broken Neural Scaling Laws. In *International Conference on Learning Representations (ICLR)*, 2023.
- [16] Vivien Cabannes, Elvis Dohmatob, and Alberto Bietti. Scaling Laws for Associative Memories. In *International Conference on Learning Representations (ICLR)*, 2024.
- [17] Francesco Cagnetta, Leonardo Petrini, Umberto M. Tomasini, Alessandro Favero, and Matthieu Wyart. How Deep Neural Networks Learn Compositional Data: The Random Hierarchy Model. *Physical Review X*, 14(3):031001, 2024.
- [18] Francesco Cagnetta, Allan Raventós, Surya Ganguli, and Matthieu Wyart. Deriving Neural Scaling Laws from the Statistics of Natural Language. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. arXiv:2602.07488.
- [19] Abdulkadir Canatar, Blake Bordelon, and Cengiz Pehlevan. Spectral Bias and Task-Model Alignment Explain Generalization in Kernel Regression and Infinitely Wide Neural Networks. *Nature Communications*, 12:2914, 2021.
- [20] Andrea Caponnetto and Ernesto De Vito. Optimal Rates for the Regularized Least-Squares Algorithm. *Foundations of Computational Mathematics*, 7(3):331–368, 2007.
- [21] L  na  c Chizat, Edouard Oyallon, and Francis Bach. On Lazy Training in Differentiable Programming. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

- [22] Aidan Clark, Diego de Las Casas, Aurelia Guy, Arthur Mensch, Michela Paganini, Jordan Hoffmann, Bogdan Damoc, Blake Hechtman, Trevor Cai, Sebastian Borgeaud, et al. Unified Scaling Laws for Routed Language Models. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, 2022.
- [23] Bertrand S. Clarke and Andrew R. Barron. Information-Theoretic Asymptotics of Bayes Methods. *IEEE Transactions on Information Theory*, 36(3):453–471, 1990.
- [24] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2 edition, 2006.
- [25] Hugo Cui, Bruno Loureiro, Florent Krzakala, and Lenka Zdeborová. Generalization Error Rates in Kernel Regression: The Crossover from the Noiseless to Noisy Regime. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 10131–10143, 2021.
- [26] Amit Daniely and Gal Vardi. From Local Pseudorandom Generators to Hardness of Learning. In *Proceedings of the 34th Conference on Learning Theory (COLT)*, volume 134 of *Proceedings of Machine Learning Research*, pages 1358–1394, 2021.
- [27] Łukasz Dębowski. The Relaxed Hilberg Conjecture: A Review and New Experimental Support. *Journal of Quantitative Linguistics*, 22(4):311–337, 2015.
- [28] Łukasz Dębowski. *Information Theory Meets Power Laws: Stochastic Processes and Language Models*. Wiley, 2021.
- [29] Leonardo DeFilippis, Bruno Loureiro, and Theodor Misiakiewicz. Dimension-Free Deterministic Equivalents and Scaling Laws for Random Feature Regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- [30] Ronald DeVore, Boris Hanin, and Guergana Petrova. Neural Network Approximation. *Acta Numerica*, 30:327–444, 2021.
- [31] Ronald A. DeVore, Ralph Howard, and Charles Micchelli. Optimal Nonlinear Approximation. *Manuscripta Mathematica*, 63(4):469–478, 1989.
- [32] Elvis Dohmatob, Yunzhen Feng, Pu Yang, François Charton, and Julia Kempe. A Tale of Tails: Model Collapse as a Change of Scaling Laws. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 11165–11197, 2024.
- [33] Werner Ebeling and Thorsten Pöschel. Entropy and Long-Range Correlations in Literary English. *Europhysics Letters*, 26(4):241–246, 1994.
- [34] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, et al. Toy Models of Superposition. Transformer Circuits Thread, 2022.
- [35] Tom Henighan, Jared Kaplan, Mor Katz, Mark Chen, Christopher Hesse, Jacob Jackson, Heewoo Jun, Tom B. Brown, Prafulla Dhariwal, Scott Gray, Chris Hallacy, Benjamin Mann, Alec Radford, Aditya Ramesh, Nick Ryder, Daniel M. Ziegler, John Schulman, Dario Amodei, and Sam McCandlish. Scaling Laws for Autoregressive Generative Modeling. *arXiv preprint arXiv:2010.14701*, 2020.
- [36] Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md. Mostofa Ali Patwary, Yang Yang, and Yanqi Zhou. Deep Learning Scaling Is Predictable, Empirically. *arXiv preprint arXiv:1712.00409*, 2017.

- [37] Wolfgang Hilberg. Der bekannte Grenzwert der redundanzfreien Information in Texten—eine Fehlinterpretation der Shannonschen Experimente? *Frequenz*, 44(9–10):243–248, 1990.
- [38] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training Compute-Optimal Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 30016–30030, 2022.
- [39] Shengding Hu, Xin Liu, Xu Han, Xinrong Zhang, Chaoqun He, Weilin Zhao, Yankai Lin, Ning Ding, Zebin Ou, Guoyang Zeng, Zhiyuan Liu, and Maosong Sun. Predicting Emergent Abilities with Infinite Resolution Evaluation. In *International Conference on Learning Representations (ICLR)*, 2024.
- [40] Marcus Hutter. Learning Curve Theory. *arXiv preprint arXiv:2102.04074*, 2021.
- [41] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural Tangent Kernel: Convergence and Generalization in Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, pages 8571–8580, 2018.
- [42] Hong Jun Jeon and Benjamin Van Roy. An Information-Theoretic Analysis of Compute-Optimal Neural Scaling Laws. *arXiv preprint arXiv:2212.01365*, 2022.
- [43] Hong Jun Jeon and Benjamin Van Roy. Information-Theoretic Foundations for Neural Scaling Laws. *arXiv preprint arXiv:2407.01456*, 2024.
- [44] William B. Johnson and Joram Lindenstrauss. Extensions of Lipschitz Mappings into a Hilbert Space. In *Conference in Modern Analysis and Probability*, volume 26 of *Contemporary Mathematics*, pages 189–206. American Mathematical Society, 1984.
- [45] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling Laws for Neural Language Models. *arXiv preprint arXiv:2001.08361*, 2020.
- [46] Michael Kearns. Efficient Noise-Tolerant Learning from Statistical Queries. *Journal of the ACM*, 45(6):983–1006, 1998.
- [47] Jakub Krajewski, Jan Ludziejewski, Kamil Adamczewski, Maciej Pióro, Michał Krutul, Szymon Antoniak, Kamil Ciebiera, Krystian Król, Tomasz Odrzygóźdź, Piotr Sankowski, Marek Cygan, and Sebastian Jaszczur. Scaling Laws for Fine-Grained Mixture of Experts. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- [48] Tanishq Kumar, Zachary Ankner, Benjamin F. Spector, Blake Bordelon, Niklas Muenighoff, Mansheej Paul, Cengiz Pehlevan, Christopher Ré, and Aditi Raghunathan. Scaling Laws for Precision. In *International Conference on Learning Representations (ICLR)*, 2025.
- [49] Elizaveta Levina and Peter J. Bickel. Maximum Likelihood Estimation of Intrinsic Dimension. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 17, 2004.
- [50] Henry W. Lin and Max Tegmark. Critical Behavior in Physics and Probabilistic Formal Languages. *Entropy*, 19(7):299, 2017.
- [51] Licong Lin, Jingfeng Wu, and Peter L. Bartlett. Improved Scaling Laws in Linear Regression via Data Reuse. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, 2025. arXiv:2506.08415.

- [52] Licong Lin, Jingfeng Wu, Sham M. Kakade, Peter L. Bartlett, and Jason D. Lee. Scaling Laws in Linear Regression: Compute, Parameters, and Data. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024. arXiv:2406.08466.
- [53] Alexander Maloney, Daniel A. Roberts, and James Sully. A Solvable Model of Neural Scaling Laws. *arXiv preprint arXiv:2210.16859*, 2022.
- [54] Eric J. Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. The Quantization Model of Neural Scaling. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023. arXiv:2303.13506.
- [55] Guido Montúfar, Razvan Pascanu, Kyunghyun Cho, and Yoshua Bengio. On the Number of Linear Regions of Deep Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 27, pages 2924–2932, 2014.
- [56] Niklas Muennighoff, Alexander M. Rush, Boaz Barak, Teven Le Scao, Aleksandra Piktus, Nouamane Tazi, Sampo Pyysalo, Thomas Wolf, and Colin Raffel. Scaling Data-Constrained Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- [57] Ryumei Nakada and Masaaki Imaizumi. Adaptive Approximation and Generalization of Deep Neural Network with Intrinsic Dimensionality. *Journal of Machine Learning Research*, 21(174):1–38, 2020.
- [58] Yoonsoo Nam, Nayara Fonseca, Seok Hyeong Lee, Chris Mingard, and Ard A. Louis. An Exactly Solvable Model for Emergence and Scaling Laws in the Multitask Sparse Parity Problem. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- [59] Elliot Paquette, Courtney Paquette, Lechao Xiao, and Jeffrey Pennington. 4+3 Phases of Compute-Optimal Neural Scaling Laws. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- [60] Phil Pope, Chen Zhu, Ahmed Abdelkader, Micah Goldblum, and Tom Goldstein. The Intrinsic Dimension of Images and Its Impact on Learning. In *International Conference on Learning Representations (ICLR)*, 2021.
- [61] Tomer Porian, Mitchell Wortsman, Jenia Jitsev, Ludwig Schmidt, and Yair Carmon. Resolving Discrepancies in Compute-Optimal Scaling of Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- [62] Jorma Rissanen. Stochastic Complexity and Modeling. *The Annals of Statistics*, 14(3):1080–1100, 1986.
- [63] Jorma Rissanen. Fisher Information and Stochastic Complexity. *IEEE Transactions on Information Theory*, 42(1):40–47, 1996.
- [64] Jonathan S. Rosenfeld, Amir Rosenfeld, Yonatan Belinkov, and Nir Shavit. A Constructive Prediction of the Generalization Error Across Scales. In *International Conference on Learning Representations (ICLR)*, 2020.
- [65] Yangjun Ruan, Chris J. Maddison, and Tatsunori Hashimoto. Observational Scaling Laws and the Predictability of Language Model Performance. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.

- [66] Nikhil Sardana, Jacob Portes, Sasha Doubov, and Jonathan Frankle. Beyond Chinchilla-Optimal: Accounting for Inference in Language Model Scaling Laws. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 43445–43460, 2024.
- [67] Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are Emergent Abilities of Large Language Models a Mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- [68] Johannes Schmidt-Hieber. Nonparametric Regression Using Deep Neural Networks with ReLU Activation Function. *The Annals of Statistics*, 48(4):1875–1897, 2020.
- [69] Shai Shalev-Shwartz, Ohad Shamir, and Shaked Shammah. Failures of Gradient-Based Deep Learning. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *Proceedings of Machine Learning Research*, pages 3067–3075, 2017.
- [70] Claude E. Shannon. A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3–4):379–423, 623–656, 1948.
- [71] Claude E. Shannon. Prediction and Entropy of Printed English. *Bell System Technical Journal*, 30(1):50–64, 1951.
- [72] Utkarsh Sharma and Jared Kaplan. A Neural Scaling Law from the Dimension of the Data Manifold. *arXiv preprint arXiv:2004.10802*, 2020.
- [73] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI Models Collapse when Trained on Recursively Generated Data. *Nature*, 631:755–759, 2024.
- [74] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM Test-Time Compute Optimally Can Be More Effective than Scaling Model Parameters. In *International Conference on Learning Representations (ICLR)*, 2025.
- [75] Ben Sorscher, Robert Geirhos, Shashank Shekhar, Surya Ganguli, and Ari S. Morcos. Beyond Neural Scaling Laws: Beating Power Law Scaling via Data Pruning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 19523–19536, 2022.
- [76] Stefano Spigler, Mario Geiger, and Matthieu Wyart. Asymptotic Learning Curves of Kernel Methods: Empirical Data versus Teacher–Student Paradigm. *Journal of Statistical Mechanics: Theory and Experiment*, 2020(12):124001, 2020.
- [77] Ryosuke Takahira, Kumiko Tanaka-Ishii, and Łukasz Dębowski. Entropy Rate Estimates for Natural Language—A New Extrapolation of Compressed Large-Scale Corpora. *Entropy*, 18(10):364, 2016.
- [78] Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer Series in Statistics. Springer, 2009.
- [79] Pablo Villalobos, Anson Ho, Jaime Sevilla, Tamay Besiroglu, Lennart Heim, and Marius Hobbhahn. Will We Run Out of Data? Limits of LLM Scaling Based on Human-Generated Data. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- [80] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent Abilities of Large Language Models. *Transactions on Machine Learning Research (TMLR)*, 2022.

- [81] Yuhong Yang and Andrew Barron. Information-Theoretic Determination of Minimax Rates of Convergence. *The Annals of Statistics*, 27(5):1564–1599, 1999.
- [82] Dmitry Yarotsky. Optimal Approximation of Continuous Functions by Very Deep ReLU Networks. In *Proceedings of the 31st Conference on Learning Theory (COLT)*, volume 75 of *Proceedings of Machine Learning Research*, pages 639–649, 2018.
- [83] George Kingsley Zipf. *Human Behavior and the Principle of Least Effort: An Introduction to Human Ecology*. Addison-Wesley Press, 1949.